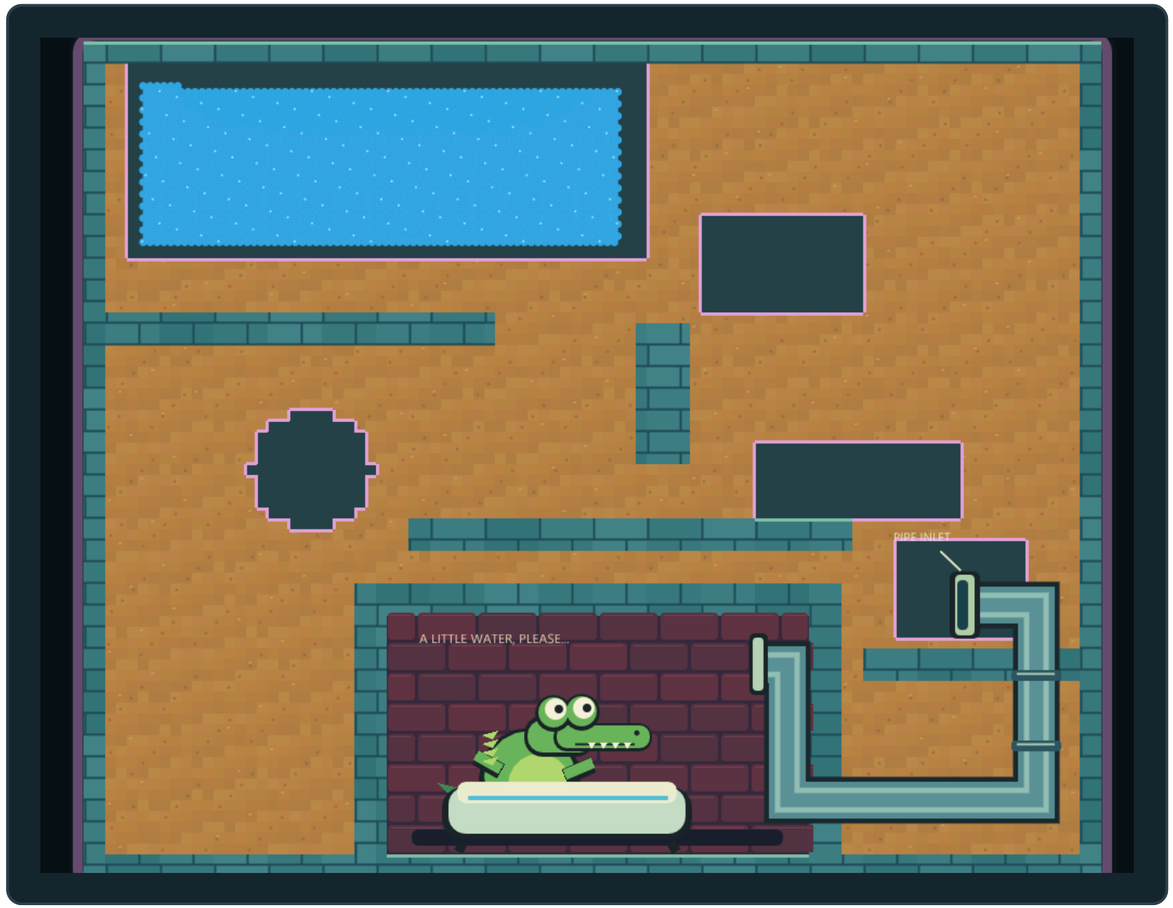
HW3

2D PIC/FLIP Fluid Simulation

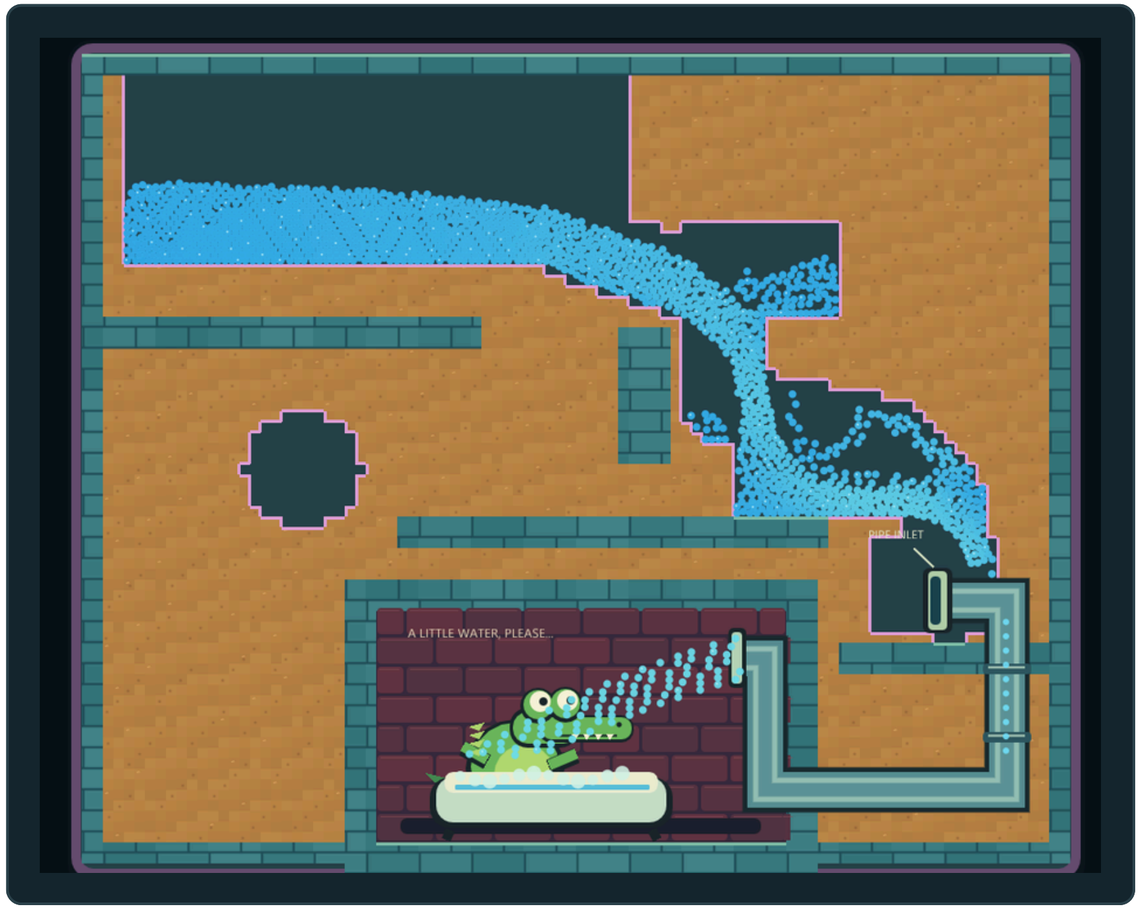
Due: 2026/11/20 (Fri.) 23:59

What are we building?

START



RUNNING

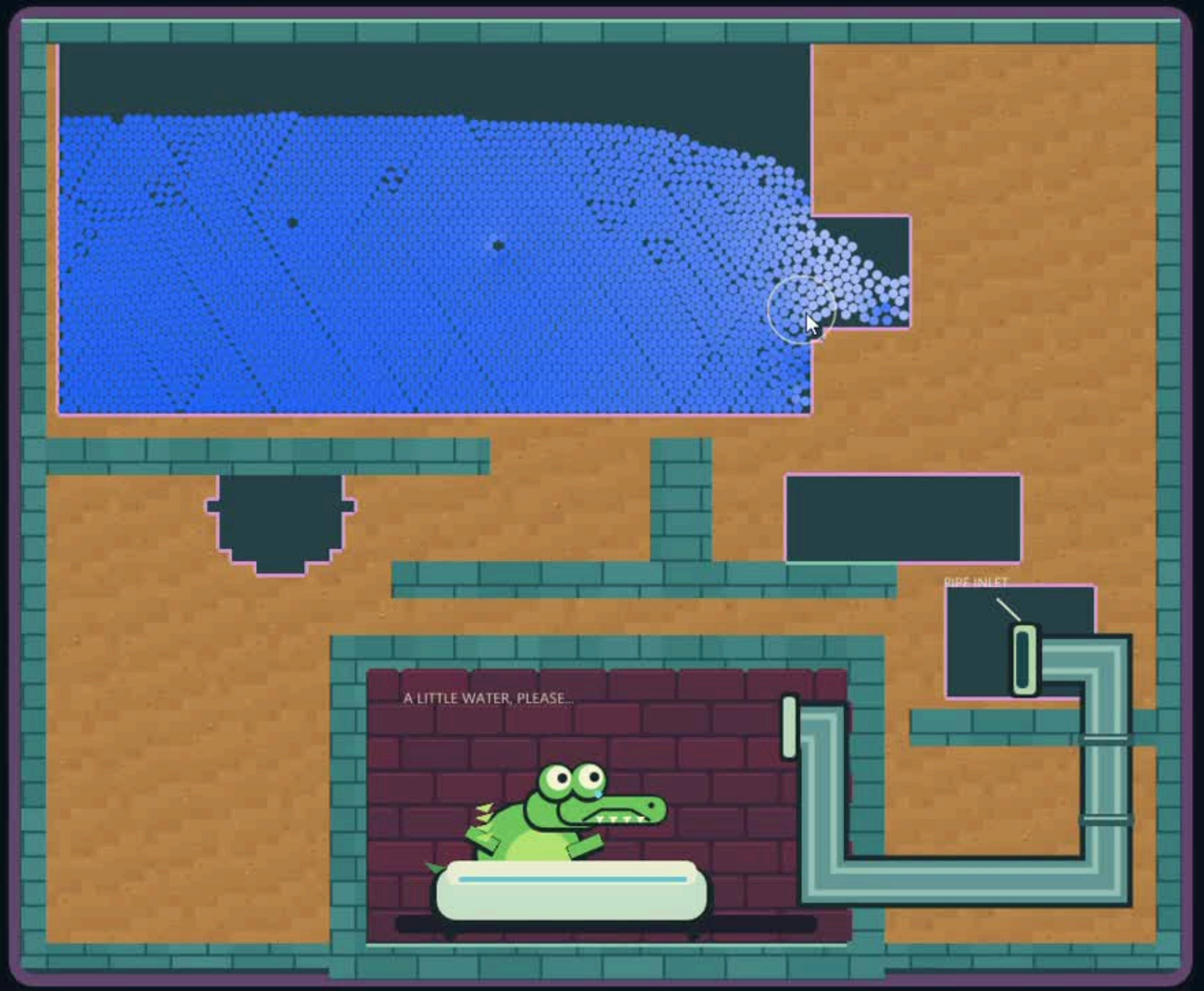


DIG ⟶ PIPE ⟶ SHOWER ⟶ BATHTUB

315551078

01 / THE UNDERGROUND BATH

DIG > PIPE > SHOWER



A LITTLE DIG. A HAPPY CROC.

Croc's Waterworks

PIC / FLIP - interactive fluid puzzle

Carve a path through the soil. Guide the water into the pipe on the right to give our little croc a bath.

01 MAKE A PATH

DIG MODE

28 px Brush

Drag to carve. Scroll to resize.

Teal stone walls are permanent. Dig around them to reach the pipe inlet.

02 FILL THE BATH

Collected water 0%

Water collected 0 / 2600

Clear threshold: 70% Great: 85% Perfect: 95%

In pipe 0 In scene 2600

70% clear / 85% great / 95% perfect

Pause [SPACE] Restart [R]

▶ Simulation settings

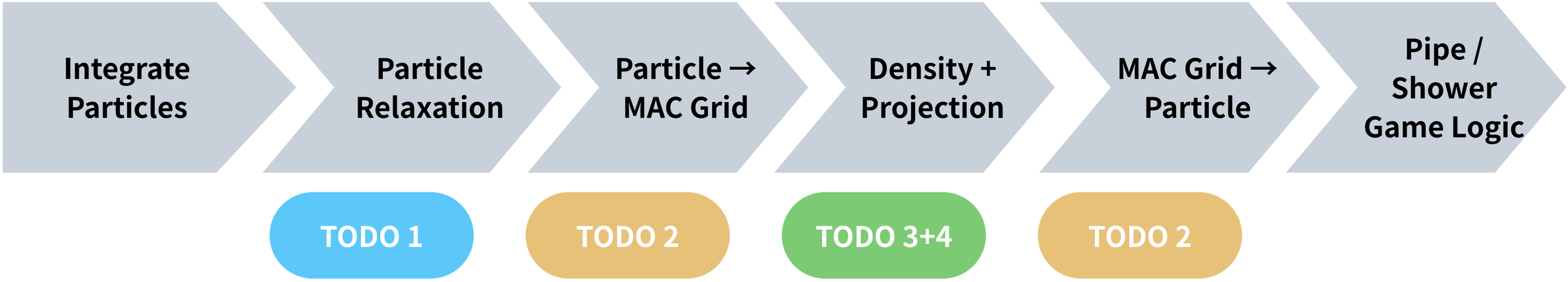
What you implement vs. what we provide

Four core fluid algorithms

- TODO 1 — Particle Relaxation
- TODO 2 — PIC/FLIP Velocity Transfer + Face Validity
- TODO 3 — Cell Density Estimation
- TODO 4 — Incompressibility Solver

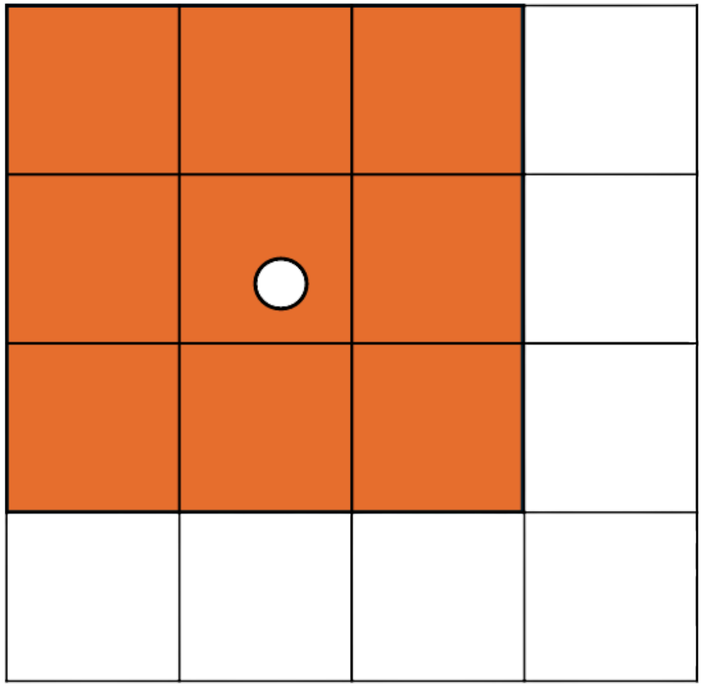
Everything around the solver

- CMake / OpenGL / ImGui / rendering
- Interactive terrain digging + fixed walls
- Pipe → shower transport + bathtub scoring
- Crocodile feedback + level UI
- Scene setup, collisions, and update pipeline

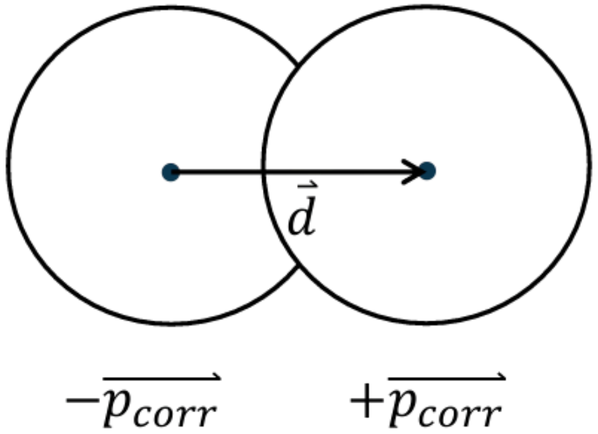


TODO 1

Particle Relaxation



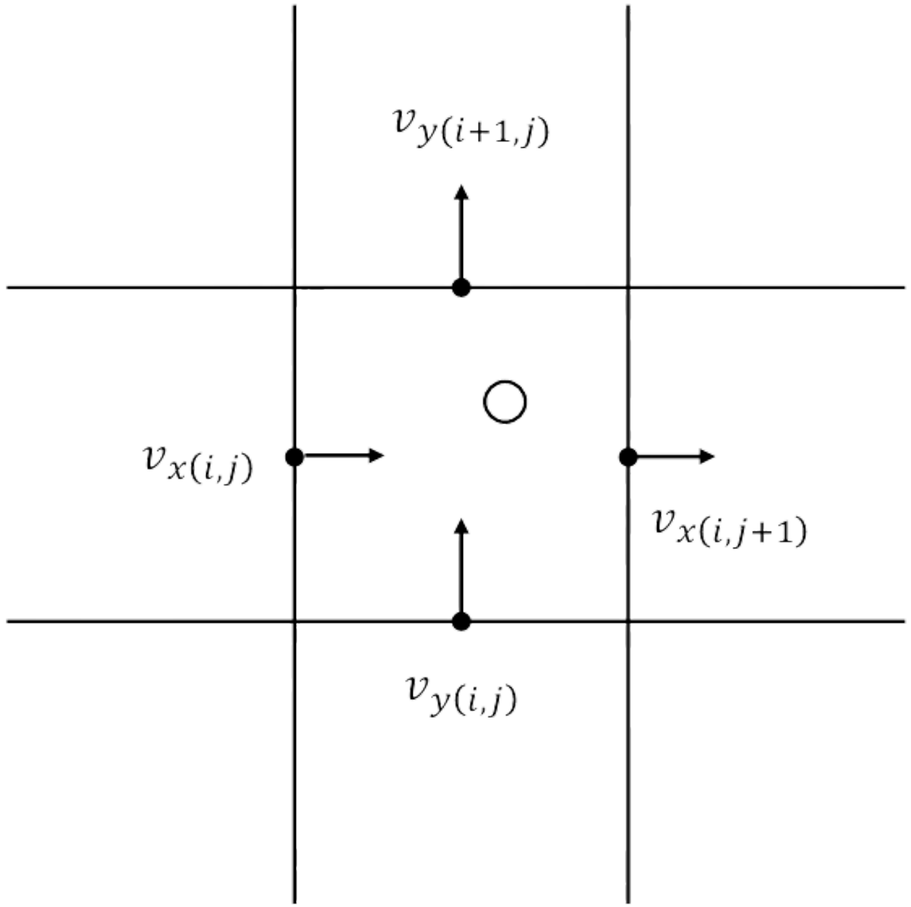
3x3 Local Lookup



$$\overrightarrow{p_{corr}} = \frac{1}{2} * \frac{2r - |\vec{d}|}{|\vec{d}|} * \vec{d}$$

TODO 2

MAC Grid



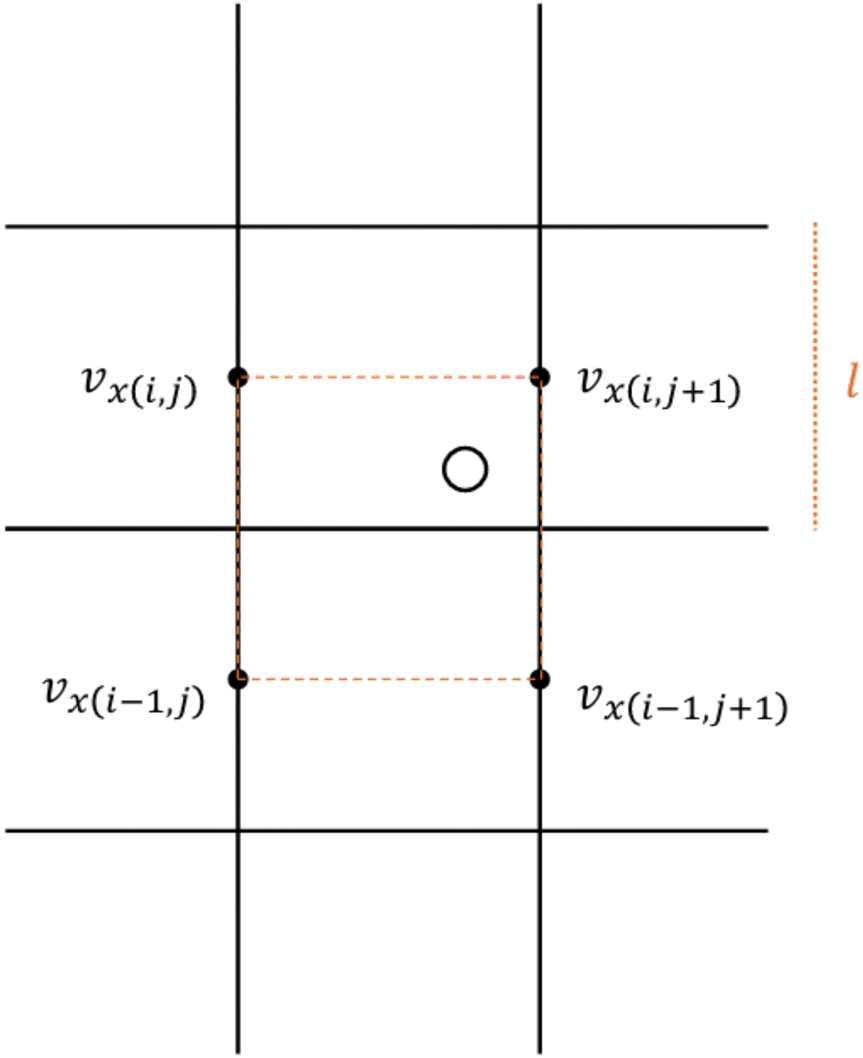
Fluid Cells

Air Cells

Solid Cells

TODO 2

Bilinear Interpolation on MAC Grid

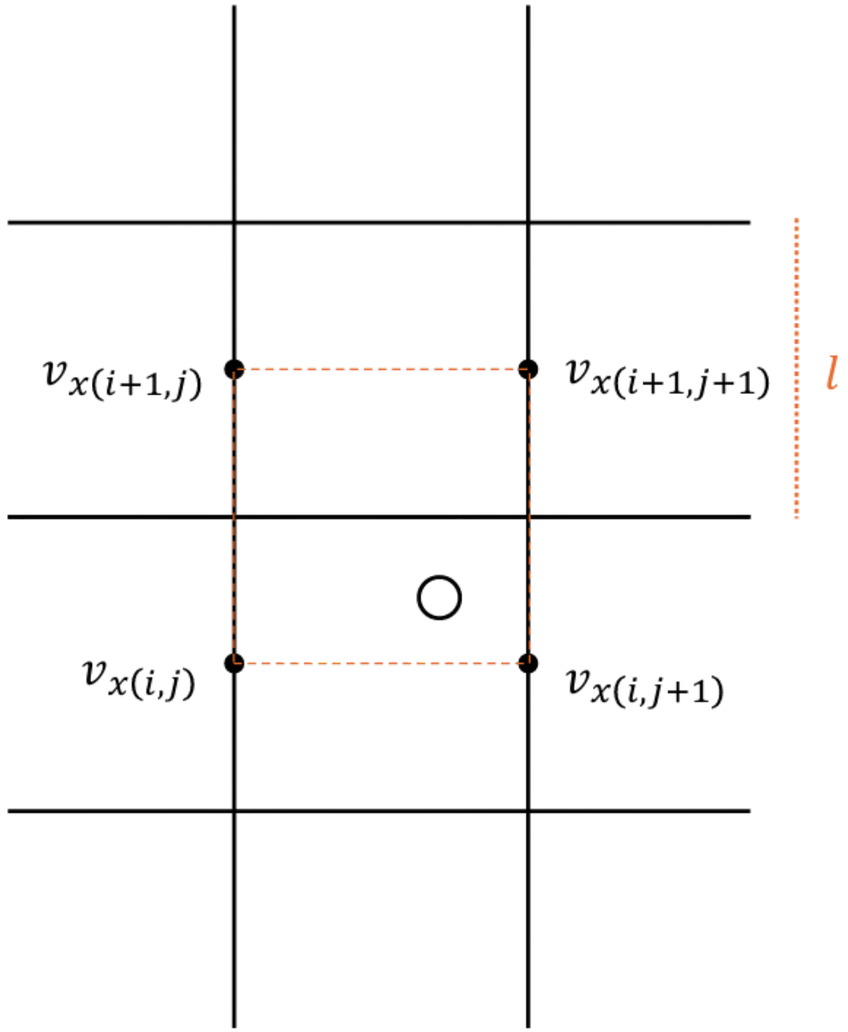


$$(i,j) = (\lfloor \frac{y - \frac{l}{2}}{l} \rfloor, \lfloor \frac{x}{l} \rfloor)$$

Lower Cell Coordinate

TODO 2

Bilinear Interpolation on MAC Grid

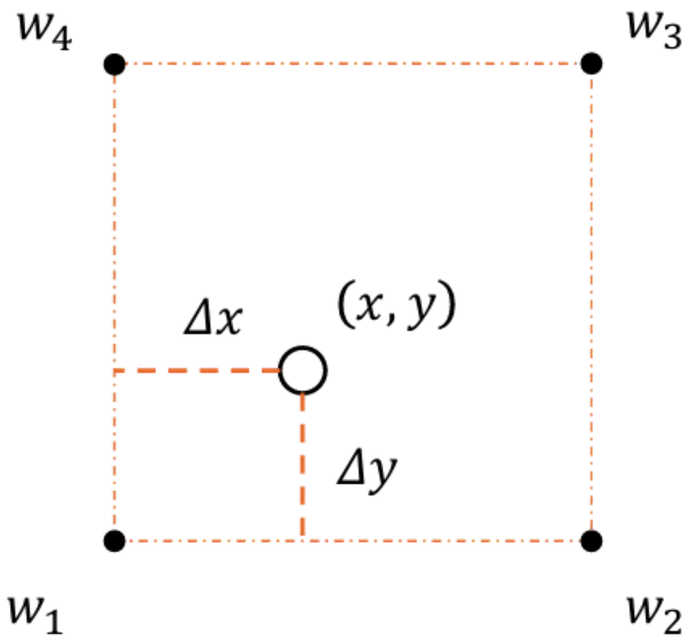


$$(i,j) = (\lfloor \frac{y - \frac{l}{2}}{l} \rfloor, \lfloor \frac{x}{l} \rfloor)$$

Lower Cell Coordinate

TODO 2

Bilinear Interpolation on MAC Grid



$$\Delta x = x - \lfloor \frac{x_{sample}}{l} \rfloor \qquad \Delta y = y - \lfloor \frac{y_{sample}}{l} \rfloor$$

$$w_1 = \left(1 - \frac{\Delta x}{l}\right)\left(1 - \frac{\Delta y}{l}\right)$$

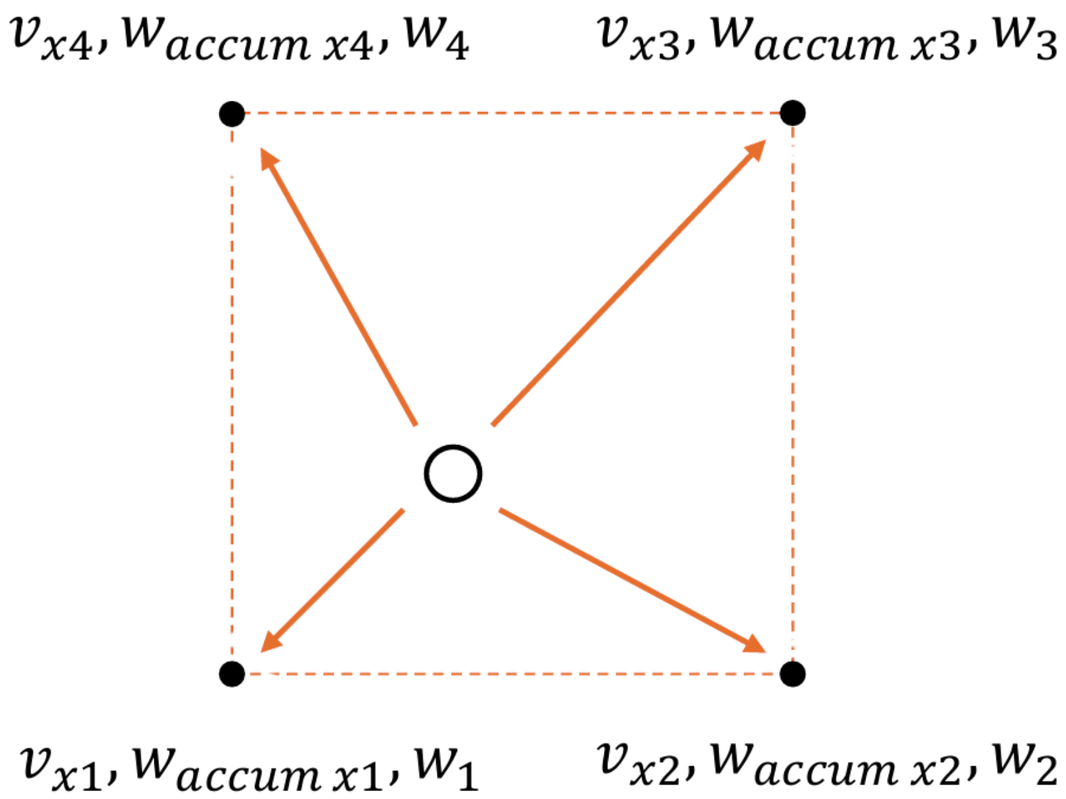
$$w_2 = \left(\frac{\Delta x}{l}\right)\left(1 - \frac{\Delta y}{l}\right)$$

$$w_3 = \left(\frac{\Delta x}{l}\right)\left(\frac{\Delta y}{l}\right)$$

$$w_4 = \left(1 - \frac{\Delta x}{l}\right)\left(\frac{\Delta y}{l}\right)$$

TODO 2

Particle Velocities to Cells



For each particle

$$v_{xn} \mathrel{+}= v_{px} * w_n$$

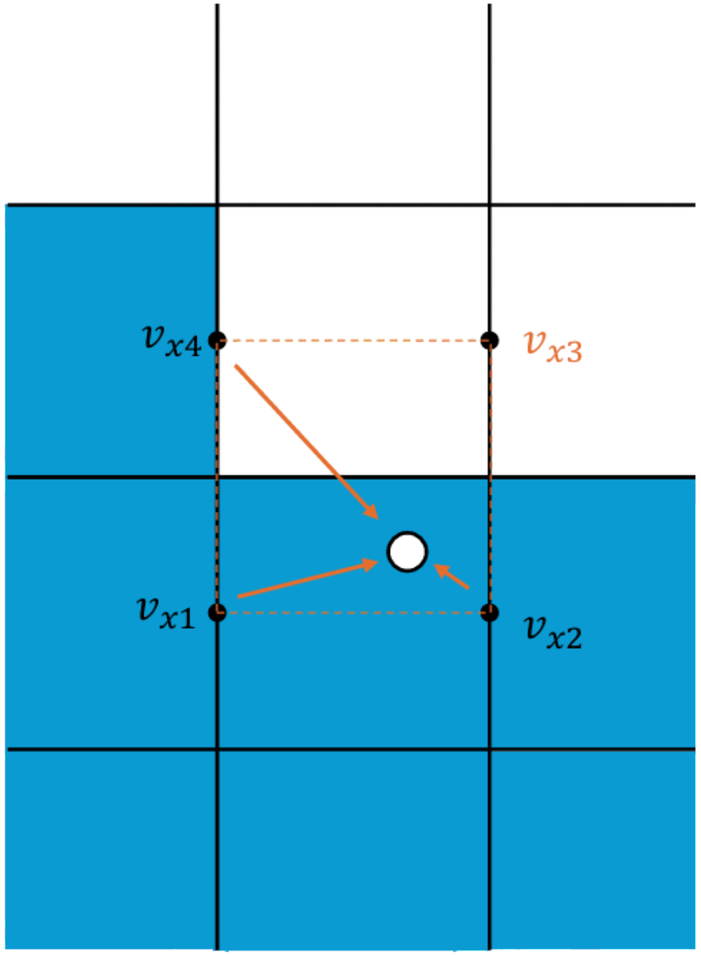
$$w_{accum\ n} \mathrel{+}= w_n$$

For each cell

$$v_{xn} = \frac{v_{xn}}{w_{accum\ xn}}$$

TODO 2

Cell Velocities to Particles



v_{x3} is invalid

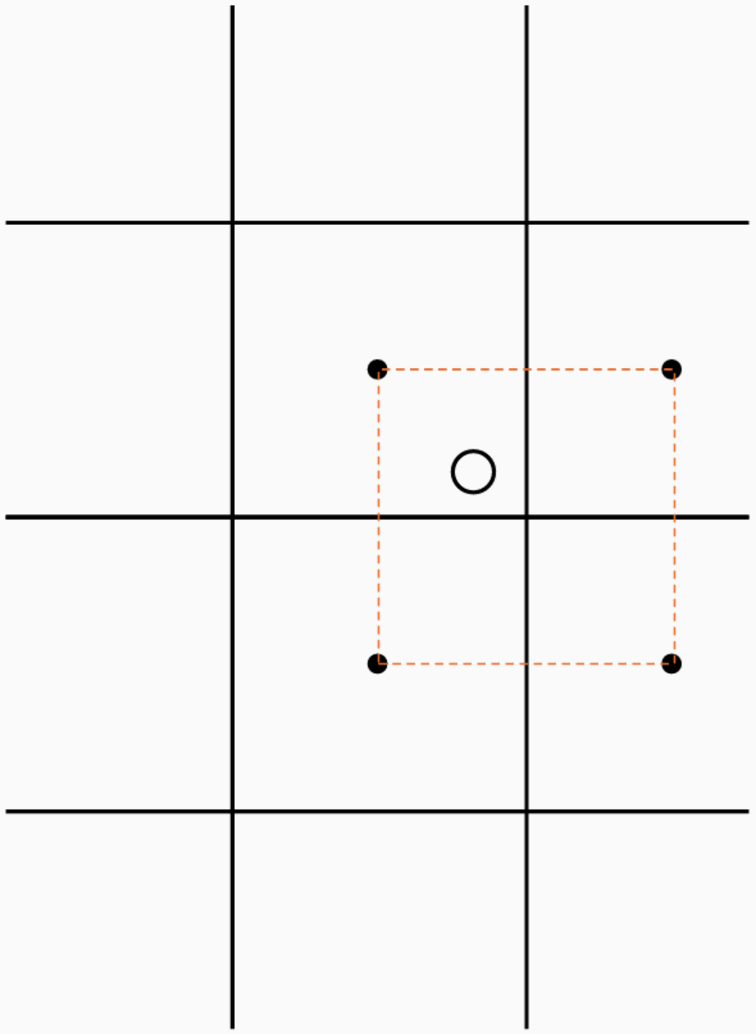
$$v_{pic\,x} = \frac{w_1 v_{x1} + w_2 v_{x2} + w_4 v_{x4}}{w_1 + w_2 + w_4}$$

$$v_{flip\,x} = v_{px} + \frac{w_1(v_{x1} - v_{xprev1}) + w_2(v_{x2} - v_{xprev4}) + w_4(v_{x4} - v_{xprev4})}{w_1 + w_2 + w_4}$$

$$v_{px} = \alpha_{flip} v_{flipx} + (1 - \alpha_{flip}) v_{picx}$$

TODO 3

Cell Density

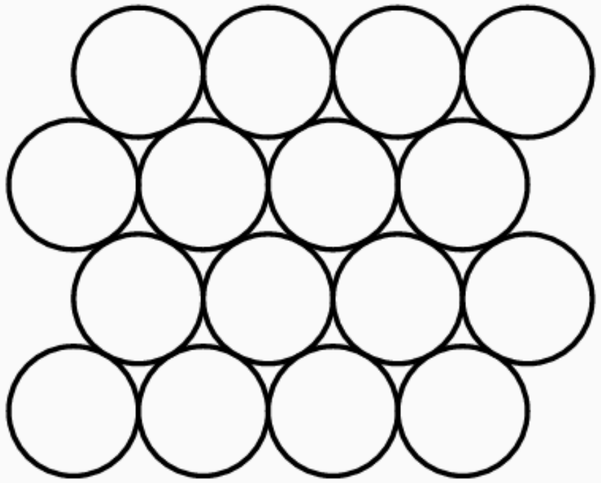


Sample cell densities at the center of each cell

Add $1 * w_n$ to each sample corner

TODO 3

Rest Density and Density Correction



Initial particle position

Calculate resting density of fluid cells

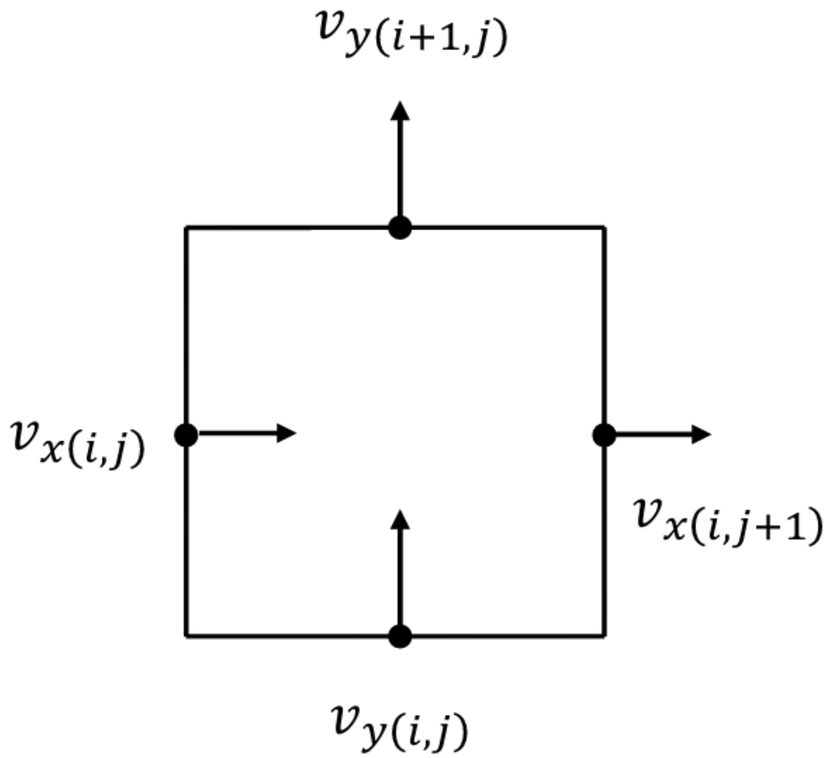
$$\rho_{rest} = \frac{Sum\ of\ fluid\ cell\ densities}{N_{fluid\ cells}}$$

Correct divergence when $\rho_{cell} > \rho_{rest}$

$$div\ -= k_{stiff}(\rho_{cell} - \rho_{rest})$$

TODO 4

Solving Incompressibility



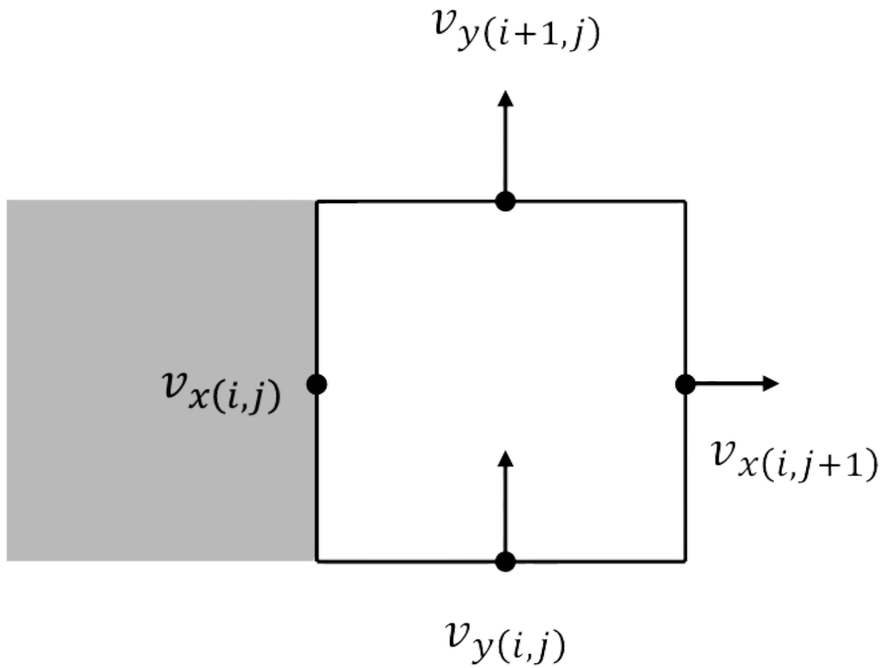
Total inflow should equal total outflow
→ Solve for divergence = 0

$$div = v_{top} + v_{right} - v_{left} - v_{bottom}$$

Bottom and left velocities are inverted when calculating divergence

TODO 4

Solving Incompressibility



Divergence correction should be distributed across available flow directions only

$$v_{corr} = o * \frac{div}{dirs}$$

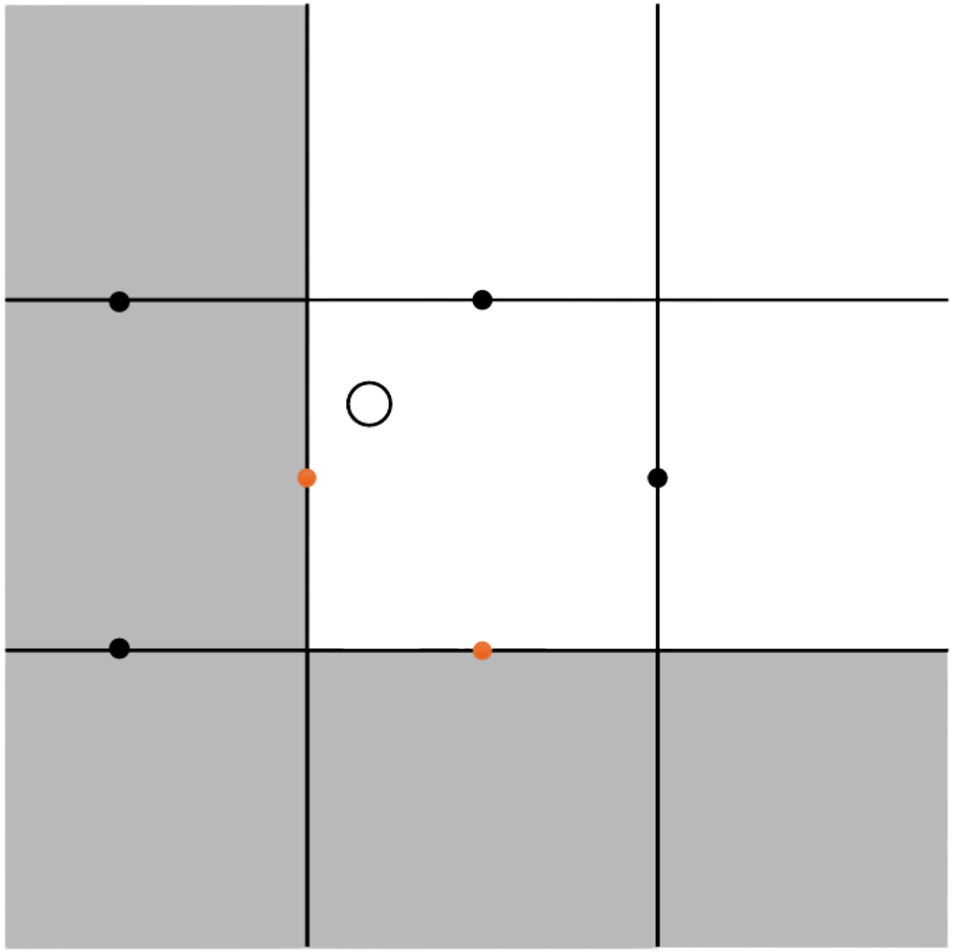
o: Over relaxation constant to speed up convergence

$$1 < o < 2$$

Add correction to the left and bottom velocities
Deduct correction from the top and right velocities
(In this case, we shouldn't add anything to the left since it is a solid cell)

TODO 2

Particle Velocities to Cells



If we set the cell velocity component to zero
→ Particle sticks to ceiling

Remove solid cells on the ceiling or we can keep its velocity but treat it as zero when solving incompressibility

Directly modify particle velocities instead when colliding with outer edges (already implemented for you)

Build, run, and debug

BUILD

Windows 10/11 is the officially supported environment.

```
.\build.bat
.\run.bat
```

Manual CMake

```
cmake -S . -B build
cmake --build build --config Release
```

Before building:
replace YOUR_STUDENT_ID in main.cpp.

CONTROLS

- Left mouse drag — dig terrain
- Mouse wheel — change dig radius
- Space — pause / resume
- R — restart level
- 1 / 2 / 3 — particles / grid / both

DEBUG ORDER

1. Relaxation 2. P→G 3. Density
4. Projection 5. G→P

```
Optional: ctest --test-dir build -C Release
--output-on-failure
```

Report

- **Implementation**
  - Explain the reasoning behind your implementation of TODO 1–4.
- **Results & Discussion**
  - Complete the three required parameter experiments:
    - FLIP Ratio: 0.0 / 0.5 / 0.95
    - Solver Iterations: 1 / 3 / 8
    - Stiffness Coefficient: 0.0 / 1.0 / 3.0
  - Change only one parameter at a time and keep all other parameters at their default values.
- **Final Demo**
  - Include at least one screenshot showing that water can successfully reach the bathtub.
- **Bonus / Miscellaneous (Optional)**
  - Briefly describe any creative extension or changes outside the required TODO sections.

- **Please refer to REPORT_TEMPLATE.md for the report format and detailed requirements.**

Scoring

- TODO 1 — particleRelaxation (15%)
- TODO 2 — transferVelocities + isFaceVelocityValid (25%)
  - Face validity (5%)
  - Particle To Cell (10%)
  - Cell To Particle / PIC-FLIP (10%)
- TODO 3 — updateDensity & density correction (15%)
  - Cell densities (10%)
  - Resting density (2%)
  - Density correction (3%)
- TODO 4 — solveIncompressibility (25%)
- Report (20%)
- Bonus (Up to +15%)

Submission

- Include all files that you have modified.
- Construct the submission folder following the project structure below.
- Do not include build/, executables, or CMake cache files.

```
HW3_<studentID>.zip/
├── main.cpp
├── src/
│   └── fluid.cpp
└── report_<studentID>.pdf
```

- Deadline
  - **2026/11/20 (Fri.) 23:59**
- Late policies
  - Penalty of 10 points on each day after deadline
- Cheating policies
  - 0 points for any cheating on assignments

Good luck with HW3!